MA378 Chapter 3: Numerical Integration

# §3.2 Simpson's Rule

Dr Niall Madden

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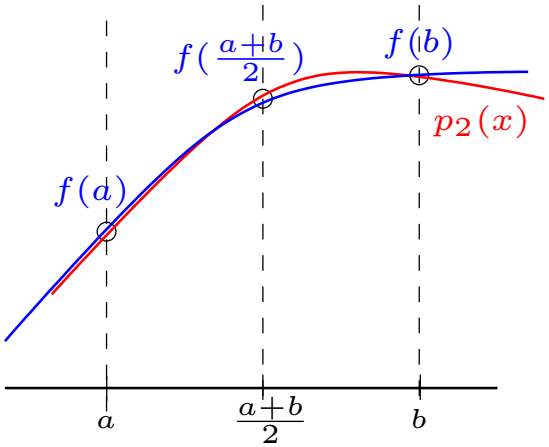


T. Simpson's Rule:

$\int_a^b f(x)dx \approx$

$\frac{b-a}{4}\left(f(a) + 4f(\frac{a+b}{2}) + f(b)\right)$

H. Simpson's Rule:

*"If something is hard to do, it is not worth doing"*.

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# 2.1 Simpson's Rule

Following on from the Trapezium Rule, we'll consider the **3-point Newton-Cotes scheme**[1]. which is based on integrating the quadratic interpolant to $f(x)$.

**Simpson's Rule**

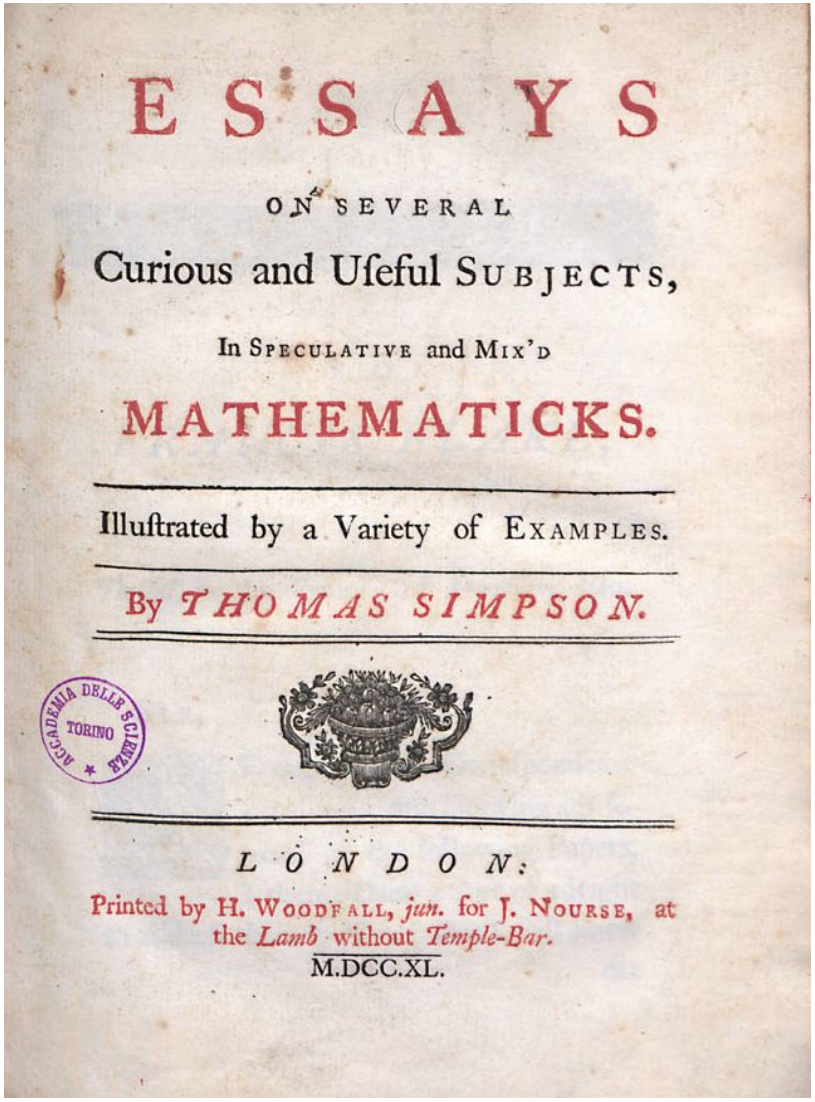
$$Q_2(f) = \frac{b-a}{6}\big(f(a) + 4f\big(\frac{a+b}{2}\big) + f(b)\big). \qquad (1)$$

[1] Thomas Simpson, 1710–1761. One of the most distinguished of a group of lecturers who taught in the London coffee-houses, Hutton (famous text-book writer) said of him

> *It has been said that Mr Simpson frequented low company, with whom he used to guzzle porter and gin: but it must be observed that the misconduct of his family put it out of his power to keep the company of gentlemen, as well as to procure better liquor.*

The method was known well before Simpson's time: it had been used by Cavalieri (a student of Galileo) in 1639, James Gregory, Johannes Kepler, and others.

# 2.1 Simpson's Rule



ESSAYS

ON SEVERAL

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Illustrated by a Variety of EXAMPLES.

By THOMAS SIMPSON.

ACCADEMIA DELLE SCIENZE TORINO

LONDON:

Printed by H. WOODFALL, *jun.* for J. NOURSE, at the *Lamb* without *Temple-Bar*.

M.DCC.XL.

Simpson does seem to have been a colourful character... He eventually appointed head of mathematics at the Royal Military Academy at Woolwich, and had a major impact on the introduction of calculus to the curriculum. His text-books were quite famous (and controversial) at the time.

He's most famous for "Simpson's Rule", which he did not invent: he learned it from Newton. But Newton got credited with the modern form of the Newton-Raphson method, which was devised by Simpson.

## 2.2 Derivation

To show how to derive Simpson's Rule, we'll use the **Method of Undetermined Coefficients** again.

First restrict our attention to approximating $\int_0^1 g(x)dx$. This method should be exact for all constant, linear and quadratic polynomials. Taking

$$g(x) \equiv 1, \qquad g(x) = x \quad \text{and} \qquad g(x) = x^2,$$

we get the set of equations:

## 2.2 Derivation

This is easily solved giving

$$\int_0^1 g(x)dx \approx \frac{1}{6}g(0) + \frac{2}{3}g(1/2) + \frac{1}{6}g(1). \tag{2}$$

To extend this to the interval $[a, b]$, we again use a change of variables to get the general Simpson's Rule (1) .

# 2.2 Derivation

**Example 2.1**

Use Simpson's rule to estimate $\int_0^{\pi/4} \cos(x)dx$, and calculate the (exact) error $|\int_a^b f(x)dx - Q_2(f)|$.

# 2.3 Newton-Cotes error estimates

We'll now derive error estimates for general Newton-Cotes methods, and look at the specific cases of the Trapezium and Simpson's rules.

**Theorem 2.2**

*Let* $M_{n+1} := \max_{a \le x \le b} |f^{(n+1)}(x)|$, *and* $\pi_{n+1}(x)$ *be the usual nodal polynomial. Define*

$$\mathcal{E}_n := \Big| \int_a^b f(x)dx - Q_n(f) \Big|.$$

*Then*

$$\mathcal{E}_n \le \frac{M_{n+1}}{(n+1)!} \int_a^b |\pi_{n+1}(x)| dx,$$

The proof just comes directly Cauchy's Theorem.

# 2.4 Error estimates for the Trapezium Rule

**Theorem 2.3**

*For the* **Trapezium Rule**, $Q_1$,

$$\mathcal{E}_1 \leq \frac{(b-a)^3}{12} M_2. \tag{3}$$

The proof is an exercise.

**Example**

Use (3) to get an upper bound on the error for the estimate of $\int_0^{\pi/4} \cos(x)dx$ using the Trapezium rule. How does this compare with the actual error?

# 2.4 Error estimates for the Trapezium Rule

## Example 2.4

If use the Trapezium Rule to estimate the integral of $x^2$ on the interval $[0, 1]$ we get

$$\int_0^1 x^2 dx = \frac{1}{3} \qquad \text{and} \qquad Q_1(x^2) = \frac{1}{2}(0+1) = \frac{1}{2}.$$

So the error is $1/6$, exactly as the theory predicts.

# 2.5 Error estimates for Simpson's rule

One can also use our theorem to show that for Simpson's Rule

$$\mathcal{E}_2 \leq \frac{(b-a)^4}{196} M_3, \tag{4}$$

but don't bother because, although correct, it is not *sharp* (that is, it is pessimistic).

**Example**

Use (4) to get an upper bound on the error for the estimate of $\int_0^{\pi/4} \cos(x)dx$ using **Simpson's** rule. How does this compare with the actual error?

# 2.5 Error estimates for Simpson's rule

**Example 2.5**

We expect Simpson's Rule to give *exactly* the right answer for integrals of constant, linear and quadratic functions. If we take $f(x) = x^3$, $a = 0$ and $b = 1$, then formula above suggests that (approx) $\mathcal{E}_2 \leq 0.03$. But ...

# 2.6 Exercises

## Exercise 2.1

Deduce the 4-point Newton-Cotes Rule for estimating the integral $\int_0^1 f(x)dx$:

$$Q_3(f) = q_0 f(x_0) + q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3).$$

Extend the rule to estimate the integral of functions over $[a, b]$.

## Exercise 2.2

Prove the error bound given for the Trapezium rule. That is, show that

$$|\int_a^b f(x)dx - Q_1(f)| := \mathcal{E}_1 \leq \frac{(b-a)^3}{12} M_2.$$